MILITARY SPECIFICATION

CABLES; COMPOSITE TELEPHONE AND TELEVISION (16 AWG, VIDEO PAIR-POLYETHYLENE INSULATED AND 19 AWG, PAPER-OR-PULP INSULATED, LEAD COVERED, AND TAPE ARMORED)

1. SCOPE

1.1 This specification covers combination telephone and video cables intended for use in transmitting voice frequency, carrier frequency timing pulses, video frequency and other electrical signals.

1.2 Classification.- The cable shall be of the following classes as specified: (See 6.2.b)

Class 1 Unarmored lead sheath cable

Class 2 Armored lead sheath cable

1.2.1 Class 1 cables.- The following cables are covered under Class 1:

Cable Designation	Video Pair	Telephone Pair
WM-116/U	2	24
WM-117/U	4	32
WM-118/U	4	51
WM-119/U	6	29
WM-120/U	12	24
WM-121/U	12	202
WM-122/U	24	3
WM-133/U	36	3

1.2.2 Class 2 cables.- The following cables are covered under Class 2:

Cable Designation	Video Pair	Telephone Pair
WM-123/U	2	24
WM-124/U	4	32
WM-125/U	4	51
WM-126/U	6	29
WM-127/U	12	24
WM-128/U	12	202
WM-129/U	24	3
WM-132/U	36	3

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or requests for proposals form a part of this specification to the extent specified herein. In case of conflict between this specification and the following documents, this specification shall govern.

SPECIFICATIONS

Federal

J-C-98	Cable and Wire, Insulated; Methods Of Sampling and Testing
L-P-390	Plastic Molding Material, Polyethylene, Low and Medium Density
QQ-W-343	Wire, Electrical and Nonelectrical Copper (Uninsulated)

Military

MIL-C-12000	Cable, Cord, and Wire Packaging and Packing Of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection By Attributes

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society For Testing Materials

B-3	Soft or Annealed Copper Wire
D-36	Methods Of Test For Softening Point Of Bituminous Material (Ring and Ball Method)
D-113	Methods Of Test For Ductility Of Bituminous Materials

(Copies of ASTM documents may be obtained from the American Society For Testing Materials, 1916 Race Street, Philadelphia 3 Pa.)

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3. REQUIREMENTS

3.1 Material.- The material for the cable covered under this specification shall be as specified herein.

3.2 Conductors

3.2.1 Type I conductor.- Each type I conductor shall be type "S" No. 16 American Wire Gauge (AWG), solid, soft or annealed copper wire, with a nominal diameter of 50.8 mils. The elongation and diameter requirements shall be in accordance with QQ-W-343 or ASTM-B-3.

3.2.2 Type II conductor.- Each type II conductor shall be type "S" No. 19 AWG, solid, soft or annealed copper wire, with a nominal diameter of 35.9 mils. The elongation and diameter requirements shall be in accordance with QQ-W-343 or ASTM-B-3.

3.2.3 Joints in conductors.- Factory joints in conductors shall be avoided as much as possible and shall not exceed one joint per conductor per length of cable. The joints shall be made to meet the requirements specified in QQ-W-343 for splices. Tensile strength of a section of the conductor, which includes the joint, shall not be less than 90 percent of the tensile strength of an adjacent section of the conductor of equal length free from joints. The resistance of a section of the conductor, not exceeding six inches in length, which includes the joint, shall not be more than 105 percent of the resistance of an adjacent section of equal length free from joints.

3.3 Insulation

3.3.1 Type I conductor.- Each type I conductor shall be insulated with an extruded covering of expanded thermoplastic polyethylene compound to form a continuous seamless tube over the conductor.

3.3.1.1 Polyethylene insulation.- The polyethylene compound prior to extruding shall comply with the requirements for type II, grade 4 of L-P-390. The compound shall be uniform in composition and commercially free from foreign matter and undispersed resin.

3.3.2 Type II conductor.- Each type II conductor shall be insulated with either a single paper tape applied helically with an overlap, wood pulp or an extruded covering of expanded thermoplastic polyethylene compound, to form a continuous seamless tube over the conductor.

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3.3.2.1 Insulating paper.- The insulating paper shall be made from clear rope stock or of a stock made from clear rope with suitable proportions of either cotton rag, chemical wood pulp or both. The paper shall be free from sizing or loading materials and shall be in accordance with Table I.

TABLE I

INSULATING PAPER

Width (Inches)	Thickness (Inches)	Thickness Variation (Inches)	Composition		
			Rope (Minimum Percent)	Wood Pulp (Maximum Percent)	Cotton
Any	0.0020	± 0.0003	40	50	Remainder
Any	0.0025	± 0.0003	40	50	Remainder
Any	0.0035	± 0.0003	25	75	---
Any	0.0050	± 0.0005	15	85	---

3.3.2.1.1 Rope stock.- Clear rope stock shall be composed of manila rope or manila rope and hemp. Jute or sisal shall not be added, but sisal, caroa, flax or fibers of similar nature may be present in clear rope stock to an extent that it does not adversely affect the characteristics of the finished cable.

3.3.2.1.2 Insulating pulp.- The wood pulp used in the insulation shall be sulfate pulp, commonly called "Kraft". The wood pulp shall be free from knots, shives and foreign material, and shall be free from residual chemicals. The chemical wood pulp shall be made by either the soda or sulfate process from spruce or from an equivalent coniferous wood. Materials treated by the sulfite process shall not be used.

3.3.2.2 Polyethylene insulation.- The polyethylene insulation shall be in accordance with Type II, Grade 4 of L-P-390. (See 3.3.1.1)

3.3.3 Polyethylene filler.- The polyethylene fillers, when used, shall consist of a glass or nylon fiber core with an extruded covering of an expanded polyethylene compound.

3.3.3.1 Core.- The core shall be a combination of continuous filaments having suitable mechanical strength.

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3.3.3.2 Polyethylene.- The polyethylene, before extruding, shall be in accordance with 3.3.1.1.

3.3.4 Core belting paper.- The core belting paper shall be made of clear rope stock, 100 percent wood pulp, or clear rope stock with suitable proportions of wood pulp, cotton rag, or both. As an alternative, paper containing fibers of caroa, flax, or sisal may be substituted for clear rope stock.

3.4 Heat damage.- Care should be exercised not to damage the polyethylene insulation or core wrapping paper by excessive heating during drying, lead sheathing or armoring operations.

3.5 Cable composition

3.5.1 Type I pairs.- Two type I insulated conductors shall be twisted together with two polyethylene fillers as specified in 3.3.3, one in each interstice to form a pair. Each pair thus formed shall be wrapped with a polyethylene tape. The polyethylene material shall be in accordance with type II, grade 4 of L-P-390. (See Figure 1) As an alternate construction two type I insulated conductors shall be twisted together. Over the twisted pair shall be extruded an expanded polyethylene belt to fill the interstices, and to provide a covering over the twisted pair with a circular cross section. (See Figure 1A)

3.5.1.1 Copper tapes.- Over the polyethylene covered pair shall be applied two soft annealed copper tapes. The inner copper tape shall be longitudinally applied and the outer copper tape shall be helically applied, the copper tape shall be continuous throughout a cable length. Joints in the copper tape shall be made in a workmanlike manner.

3.5.1.2 Crepe paper.- Over each copper taped type I pair shall be applied two tapes of natural colored crepe paper. A colored binder over the paper taped pair identifies the pair in the cable.

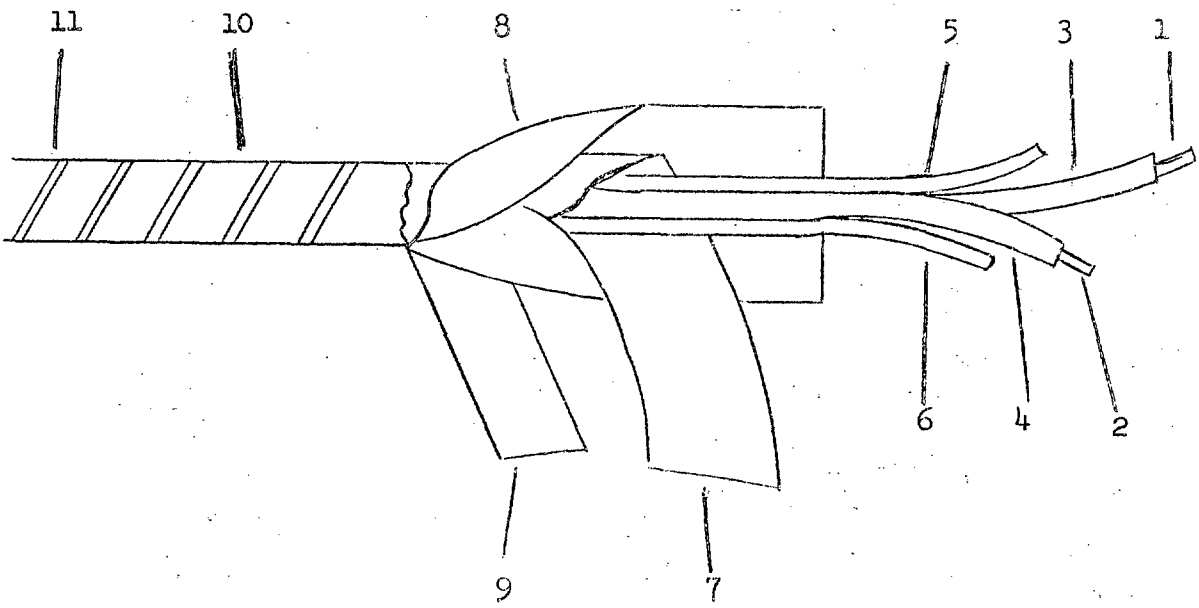
3.5.1.3 Binder colors for type I pairs.- The colors of the binder over the type I pairs shall be as specified in Table II. The binder threads shall be helically applied. The tracer threads shall be helically applied with a thread binder or placed longitudinally under the binder threads. Each pair member shall have a different length of lay. The lay system shall be devised so that adjacent and alternate pairs in a layer and pairs in adjacent layers shall have different lengths of twist.

TABLE II

COLORS OF BINDERS OF TYPE I PAIRS

Pair No.	Binder	Tracer Thread
1	red (R)	---
2	blue (B)	---
3	white (W)	black (BK)

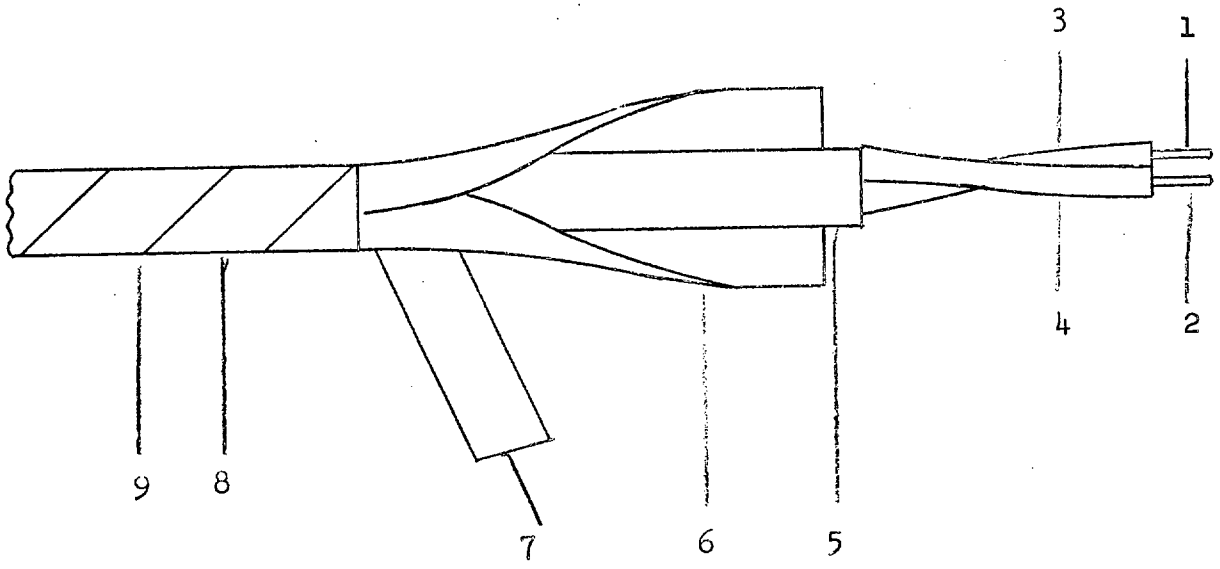
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ITEM NO.	DESCRIPTION
1 & 2	16 AWG copper conductors
3 & 4	Expanded polyethylene insulation - nominal O.D. 0.141
5 & 6	Fiber glass or nylon reinforced expanded polyethylene fillers - nominal O.D. 0.089
7	Polyethylene tape - nominal thickness 0.010, nominal O.D. 0.31
8	Inner copper tape - longitudinally applied with overlap nominal thickness 0.003
9	Outer copper tape helically applied with overlap nominal thickness 0.003, nominal O.D. 0.32
10	1 or 2 crepe paper tapes (total covering not less than 40 pound Kraft or equivalent) longitudinally applied, nominal O.D. 0.36
11	Colored identification binder (See Table II)

FIGURE 1 CONSTRUCTION DETAILS OF THE TYPE I PAIRS (VIDEO)

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ITEM NO.	DESCRIPTION
1 & 2	16 AWG copper conductors
3 & 4	Expanded polyethylene insulation - nominal O.D. 0.141
5	Expanded polyethylene belt filling the interstices nominal O.D. 0.31
6	Inner copper tape - longitudinally applied with overlap nominal thickness 0.003
7	Outer copper tape - helically applied with overlap nominal thickness 0.003, nominal O.D. 0.32
8	1 or 2 crepe paper tapes (total covering not less than 40 pound Kraft or equivalent) longitudinally applied, nominal O.D. 0.36
9	Colored identification binder (See Table II)

FIGURE 1A CONSTRUCTION DETAILS OF TYPE I PAIRS (VIDEO)

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TABLE II

COLORS OF BINDERS OF TYPE I PAIRS

Pair No.	Binder	Tracer Thread
4	white	slate (S)
5	white	---
6	red	green (G)
7	blue	green
8	white	blue
9	white	orange (O)
10	white	green

3.5.1.4 Construction details.- The construction details of the Type I pairs shall be as indicated in Figures 1 and 1A.

3.5.2 Type II pairs.- The colors of insulation used on the two conductors of a pair shall be different from each other. The pairs shall have the color insulation specified in Table III. The first color indicates the color of the insulation of the basic wire and the second color, its mate.

TABLE III

COLORS AND STRANDING OF UNITS

No. of Pairs in Unit	Center	1st Layer	Outside Layer	Tracer Pair Red-Blue
3	---	---	2 - GW	One in outside layer
12	3 - RW	---	8 - GW	One in outside layer
16	1 - BW	5 - RW	9 - GW	One in outside layer
17	1 - BW	5 - RW	10 - GW	One in outside layer
17	1 - GW	5 - BW	10 - RW	One in outside layer
16	1 - RW	5 - GW	9 - BW	One in outside layer
17	1 - RW	5 - GW	10 - BW	One in outside layer

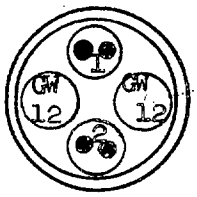
3.6 Cabling

3.6.1 Arrangement of pairs.- The general arrangement of the pairs in the cable shall be as indicated in Figures 2 and 2A.

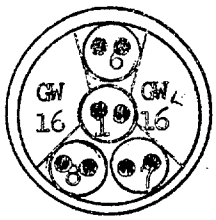
3.6.2 Type II pairs.- The Type II pairs shall be arranged in units as a unit type construction.

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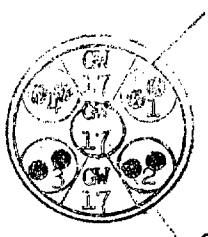
WM-116/U & WM-123/U
2 VIDEO PAIRS
24-PAIRS 19 AWG



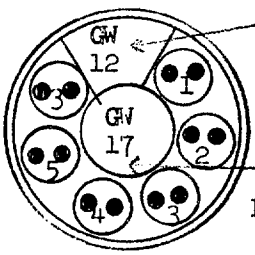
WM-117/U & WM-124/U
4 VIDEO PAIRS
32 PAIRS 19 AWG



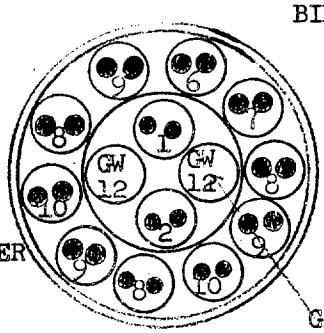
WM-118/U & WM-125/U
4 VIDEO PAIRS
51 PAIRS 19 AWG



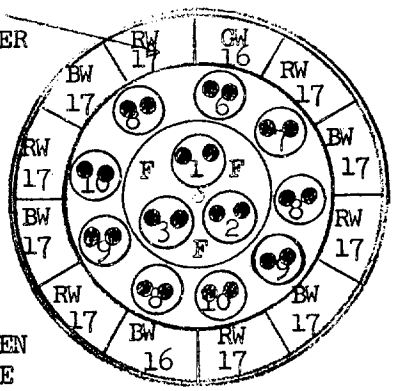
WM-119/U & WM-126/U
6 VIDEO PAIRS
29 PAIRS 19 AWG



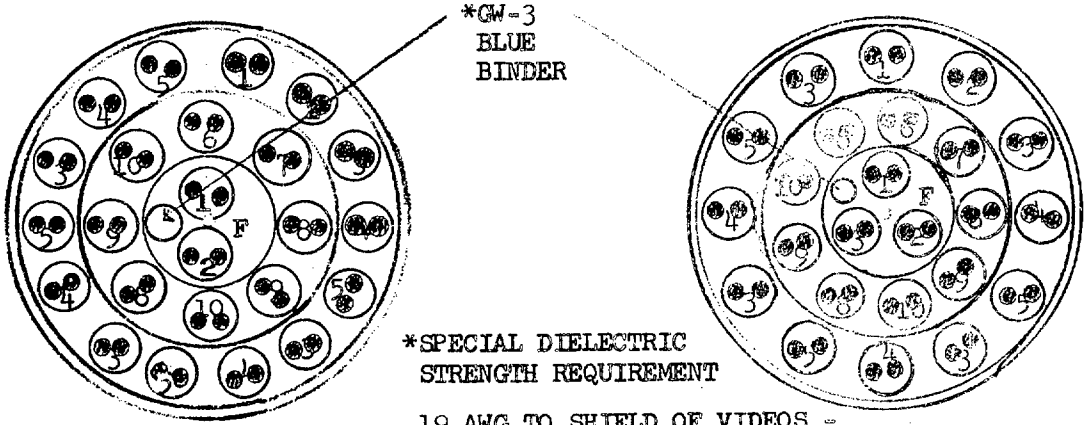
WM-120/U & WM-127/U
12 VIDEO PAIRS
24 PAIRS 19 AWG



WM-121/U & WM-128/U
12 VIDEO PAIRS
202 PAIRS 19 AWG



WM-122/U & WM-129/U
24 VIDEO PAIRS
3 PAIRS 19 AWG
ALTERNATES



*SPECIAL DIELECTRIC
STRENGTH REQUIREMENT

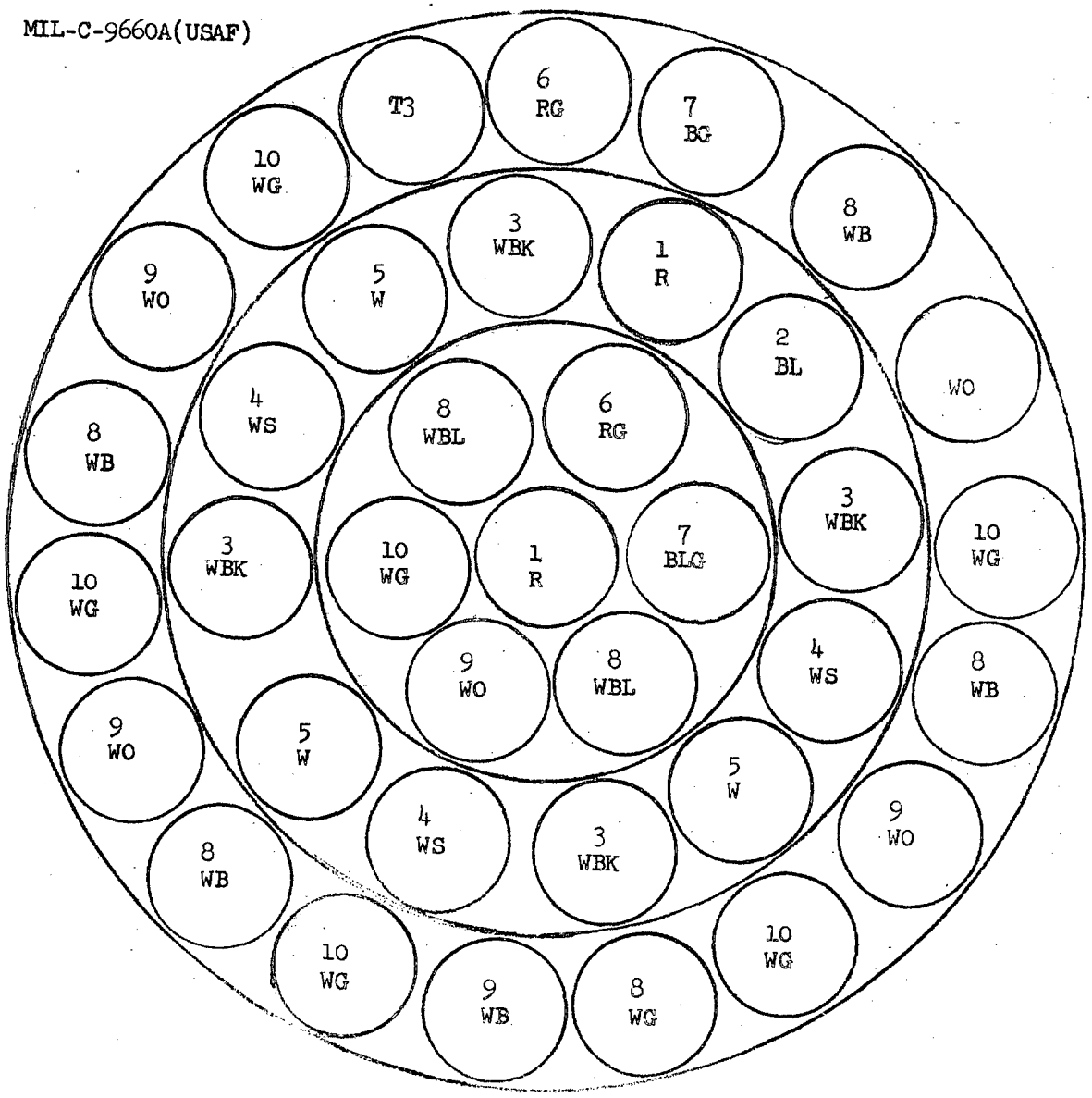
19 AWG TO SHIELD OF VIDEOS -
700 V AC POTENTIAL (2 SECS)

F - PAPER FILLERS

NOTE: SEE TABLE II FOR THE IDENTITY OF
THE VIDEO PAIRS IN THE ABOVE CABLES

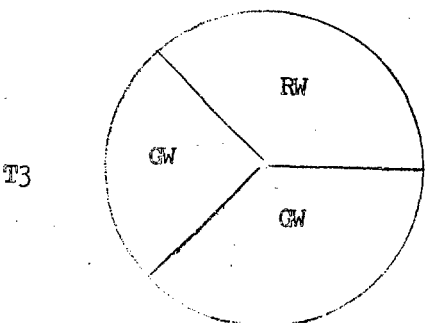
FIGURE 2 CABLING ARRANGEMENT

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WM-132/U & WM-133/U
36 VIDEO PAIRS 16 AWG AND
3 TELEPHONE PAIRS 19 AWG

FIGURE 2A CABLE ARRANGEMENT



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3.6.2.1 Unit type construction.- The colors of the pairs and the stranding of the units of 19 AWG Type II pairs shall be as specified in Table III. The size of the units and their arrangement in the cable core corresponding to the required number of Type I pairs shall be in accordance with Figure 2. The colors of the Type II units shown in Figure 2 shall be the color of the insulation of the pairs in the outside layer.

3.6.3 Core belt.- The cable core shall be covered with at least two wrappings of paper. The paper shall have a nominal thickness of not less than 0.005 inch and shall be so applied that all portions of the core are covered with not less than two thicknesses of paper.

3.6.4 "Kraft" paper fillers.- "Kraft" paper fillers shall be used when necessary in core construction to provide a concentric cable. (See Figure 2)

3.7 Lead sheath.- The core shall be inclosed in a smooth, close fitting, cylindrical lead sheath of uniform composition and thickness, free from holes or other defects. The sheath shall be applied so that the core is not subjected to nor distorted by, excessive pressure or temperature. The lead sheath shall be composed of commercially pure lead and commercially pure antimony. The amount of antimony by weight in any sample of lead sheath shall not be less than 0.9 percent nor more than 1.1 percent. The ends of the lead sheath shall be sealed, made watertight and protected from mechanical injury. The sheath with nitrogen gas introduced, shall retain a pressure of between 11 and 13 pounds per square inch (PSI).

3.7.1 Thickness of lead sheath.- The average thickness of the lead sheath, as determined by ten measurements taken at equal spacings around the circumference, shall not be less than specified in Table IV. The thickness at any point around the circumference shall not be less than 90 percent of the specified value.

3.7.2 Diameter over lead sheath.- The average diameter over the lead sheath shall not exceed the values as specified in Table IV. The average diameter shall be the average of several measurements taken with a diameter tape at equal distances along the length of the cable.

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TABLE IV

DIMENSIONAL REQUIREMENTS - LEAD SHEATH CABLES

Designation	Thickness of Lead Sheath (Inches)	Diameter Over Lead Sheath (Inches)	Size of Armor Tape Thickness		Armored Cables Over-all Diameter (Inches)
			Maximum Width (Inches)	Minimum Width (Inches)	
2 video pairs, 24 pairs 19 AWG	0.081	1.3	1.25	0.039	1.9
4 video pairs, 32 pairs 19 AWG	0.087	1.6	1.25	0.039	2.2
4 video pairs, 51 pairs 19 AWG	0.090	1.7	1.75	0.039	2.3
6 video pairs, 29 pairs 19 AWG	0.091	1.7	1.75	0.039	2.3
12 video pairs, 24 pairs 19 AWG	0.099	2.1	1.75	0.039	2.7
12 video pairs, 202 pairs 19 AWG	0.110	2.6	2.00	0.039	3.3
24 video pairs, 3 pairs 19 AWG	0.110	2.7	2.00	0.039	3.4
36 video pairs, 3 pairs 19 AWG	0.119	3.0	2.00	0.039	3.8

3.8 Protection of lead sheath.- When protection over the lead sheath is required, the lead sheath shall be flooded with a refined asphalt compound and covered with at least one impregnated paper tape not less than 0.007 inch nominal thickness, applied with overlap so as to give at least one single thickness at every point. The paper-taped cable shall then be thoroughly coated with a refined asphalt compound, after which the cable shall be served with impregnated jute roving as specified in 3.8.1 and 3.8.3 and in accordance with Table V. Where 30 or 48 pound jute roving is specified in Table V, it shall be the manufacturer's option to furnish

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either 30 or 48 pound jute. The lay of the jute roving shall not exceed four inches for cables with a diameter over the lead sheath of two inches or less, and shall not exceed five inches for cables larger than two inches in diameter. Each serving of jute roving shall be thoroughly impregnated with a refined asphalt compound application to the cable.

TABLE V

APPLICATION OF JUTE ROVING TO LEAD SHEATHED CABLE

Nominal Diameter Over Lead Sheath (Inches)	Bedding between lead sheath and tape armor (number of layers and weight)			
	Layers	Pounds	Layers	Pound
0.30 to 0.99	1	30 or 48	-	---
1.00 to 1.49	1	48	-	---
1.50 to 1.99	2	30 or 48	1	70
2.00 to 2.24	1	30 or 48	1	100
	and 1	48	-	---
2.25 and larger	2	48	1	100

3.8.1 Jute roving.- The 30, 48, 70 or 100 pound jute roving used in the cable (see Tables V and VI) shall weigh, before impregnation, 30, 48, 70 or 100 pounds, respectively, per 14,400 yards with a tolerance of ± 10 percent when corrected to 13.75 percent moisture regain, and shall be capable of supporting a minimum weight equivalent to four miles of itself.

TABLE VI

JUTE COVERING OVER TAPE ARMORED CABLE

Nominal Diameter Over Lead Sheath of Cable to be covered (Inches)	Number of layers and weight of Jute roving and jute yarn (Over armor, see 3.8.1 and 3.8.2)
0.30 to 0.99	1 layer 48 pound jute roving
1.00 to 1.49	2 layers 30 or 48 pound jute roving
1.50 to 1.74	1 layer 30 or 48 pound jute roving plus 1 layer 48 pound jute roving or 12/3 ply jute yarn

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TABLE VI

JUTE COVERING OVER TAPE ARMORED CABLE

Nominal Diameter Over Lead Sheath of Cable to be covered (Inches)	Number of layers and weight of Jute roving and jute yarn (Over armor, see 3.8.1 and 3.8.2)
1.75 to 2.29	1 layer 30 or 48 pound jute roving plus 12/3 ply jute yarn
2.3 and larger	1 layer 30 or 48 pound jute roving plus one layer 12/3 or 16/3 ply jute yarn

3.8.2 Jute yarn.- The three-ply jute yarn specified in 3.10 shall have individual strands weighing, before impregnation, 12 pounds (12/3 ply) or 16 pounds (16/3 ply) per 14,400 yards with a tolerance of ± 10 percent when corrected to a 13.75 moisture regain, and shall be capable of supporting a minimum weight equivalent to five miles of itself.

3.8.3 Presaturation of jute.- All jute used in the cable shall, before application to the cable, be thoroughly impregnated with a compound consisting of approximately 80 percent by weight of anthracene oil and approximately 20 percent of coal-tar pitch. The anthracene oil and coal-tar pitch shall be commercially free from dirt and other foreign materials. The amount of impregnant contained in any 100-foot sample of jute shall not be less than 30 percent nor more than 53 percent of the weight of the untreated jute when corrected to a 13.75 percent moisture regain. After the jute is applied to the cable, it shall not contain anthracene compound in such quantities as will penetrate the asphalt compound or prevent the jute from adhering to the cable.

3.8.4 Flooding compound.- The flooding compound used for flooding the cable at various stages of the armoring process shall be commercially free from dirt and other foreign materials. The compound, as it is being applied over the outer layer of jute roving and yarn, as specified in Table VI, shall meet the requirements specified in 3.8.4.1 and 3.8.4.2.

3.8.4.1 Softening point.- The softening point of the compound shall not be less than 62.5°C (145°F). (See 4.10)

3.8.4.2 Ductility.- The ductility of the compound at 25°C (77°F) shall not be less than two centimeters. (See 4.11)

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3.9 Armor of cables.- The cables shall be armored with two, non-galvanized, mill-finished steel tapes of the size specified in Table IV, in such a manner as to completely cover the underlaying material.

3.9.1 Divergence in width of tape.- The width of the steel tape shall not vary more than ± 0.125 inches.

3.9.2 Tensile strength.- The steel tape shall have a tensile strength of not less than 40,000 PSI.

3.9.3 Tape elongation.- The steel tape shall have an elongation of not less than 25 percent in two inches.

3.9.4 Inhibition of rust on tape.- The tape shall be free from rust and before being applied to the cable, shall be thoroughly coated with a closely adhering asphalt compound.

3.9.5 Method of applying tape.- The tapes shall be wound tightly around the cable in the same direction so as to form an open helix.

3.9.6 Alternate method of application.- As an alternate method of application the rust free tapes shall be flooded with a closely adhering asphalt compound directly prior to forming the tape on the cable.

3.9.7 Second tape.- The second tape shall cover the gaps between the edges of the first tape. The edges of successive terms of the tape shall not be so close together as to render the cable nonflexible.

3.9.8 Joints in tape.- Joints made in the steel tape shall be welded or mechanically made.

3.9.9 Spacing between joints.- In the finished cable, the spacing between joints made in the two layers of tape, shall not be less than one foot.

3.10 Jute covering.- Immediately after the armor is applied, each cable shall be thoroughly coated with a refined asphalt compound and shall then be served with jute roving (see 3.8.1) or jute yarn (see 3.8.2) as specified in Table VI. Where 30 pound jute roving is specified in Table VI, it shall be the manufacturer's option to furnish either 30 or 48 pound jute. The lay of the jute strands

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shall not exceed four inches for cables having a diameter over the lead sheath of two inches or less, and shall not exceed five inches for cables larger than two inches in diameter. Each serving of jute, after it is applied to the cable, shall be thoroughly impregnated with a refined asphalt compound. The outer layer shall be given a nonadhesive coating of powdered talc whiting (soapstone or chalk), so that the adjacent turns of the cable shall not stick together on the reels. The finished cable shall present a smooth and compact surface, which cannot be easily abraded.

3.11 Outside diameter.- The outside diameter of the armored cables shall not exceed the values as specified in Table IV. The diameter shall be determined by averaging a series of measurements made at several points on the cable with a diameter tape and at equal distances along the length of the cable.

3.12 Cable reels.- Unless otherwise specified the cable shall be wound on steel reels conforming to MIL-C-12000.

3.13 Electrical requirements

3.13.1 Type I pairs

3.13.1.1 Continuity of conductors.- Each Type I conductor in a length of cable shall be continuous. (See 4.7.1.2)

3.13.1.2 Conductor resistance.- The direct current (DC) resistance of each type I conductor shall not be more than 23 ohms per mile of completed cable at 15°C (60°F). (See 4.7.1.1)

3.13.1.3 Insulation resistance.- Each type I conductor tested shall have an insulation resistance of not less than 500 megohm (MEG) miles at a temperature not lower than 15°C (60°F), when measured against all other conductors connected to the sheath and grounded, the measurement being taken after an electrification of not more than one minute with a minimum DC potential of 500 volts. (See 4.7.1.8)

3.13.1.4 Capacitance.- The mutual capacitance (the capacitance between two wires of a pair with all the other wires grounded to the sheath) shall not be greater than 0.058 microfarads (UF) nor less than 0.051 UF per mile when referred to 15°C (60°F). The capacitance shall be measured as specified in 4.7.1.3.

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3.13.1.5 Capacitance unbalance.- The capacitance unbalance of each type I pair shall not be greater than 0.50 picofarad (PF) per foot of cable when measured as specified in 4.7.1.4.

3.13.1.6 Dielectric strength.- Each type I pair shall withstand 2800 volts between conductors, between conductors and shield and 1400 volts between shield and sheath, when measured as specified in 4.7.1.5.

3.13.1.7 Internal echo.- The internal echo measured on each end of each type I pair shall not be less than 38 decibels (DB) as specified in 4.7.1.6.

3.13.1.8 Terminal impedance.- The terminal impedance of each type I pair shall be 125.5 ± 3.5 ohms when measured as specified in 4.7.1.7.

3.13.2 Type II pairs

3.13.2.1 Continuity of conductors.- Each type II conductor in a length of cable shall be continuous. (See 4.7.2.2)

3.13.2.2 Conductor resistance.- The DC resistance of each type II conductor at 15°C (60°F) in a length of cable shall not be more than 46 ohms per mile of completed cable. (See 4.7.2.1)

3.13.2.3 Insulation resistance.- Each type II conductor tested shall have an insulation resistance of not less than 500 megohm miles at a temperature not lower than 15°C (60°F), when measured against all other conductors connected to the sheath and grounded, the measurement being taken after an electrification of not more than one minute with a minimum DC potential of 500 volts. (See 4.7.2.8)

3.13.2.4 Capacitance.- The mutual capacitance of type II pairs shall be 0.083 UF per mile. The average mutual capacitance of type II pairs in any cable shall not exceed 0.090 UF per mile for unarmored cable nor 0.092 UF per mile for armored cable. The mutual capacitance shall be measured at a frequency of 900 or 1000 cycles per second (CPS) ± 10 percent. (See 4.7.2.3)

3.13.2.5 Capacitance unbalance.- The capacitance unbalance of each type II pair shall not be greater than 0.50 PF per foot of cable when measured as specified in 4.7.2.4.

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3.13.2.6 Dielectric strength.- The insulation of each type II conductor shall withstand for two seconds a 60 CPS alternating-current (AC) potential of not less than 700 volts, maximum instantaneous value, when applied between each conductor in turn and the remaining conductors connected together. (See 4.7.2.5)

3.13.2.6.1 The insulation between each type II conductor and the shields of type I pairs and the lead sheath shall withstand for two seconds a 60 CPS AC potential of 1400 volts, maximum instantaneous value.

3.13.2.7 Internal echo.- The internal echo measured on each end of each type II pair shall not be less than 38 DB, as specified in 4.7.2.6.

3.13.2.8 Terminal impedance.- The terminal impedance of each type II pair shall be 125.5 $\pm$ 3.5 ohms when measured as specified in 4.7.2.7.

3.14 Defective pairs

3.14.1 Defective type I pairs.- There shall be no defective type I pairs.

3.14.2 Defective type II pairs.- There shall be no defective type II pairs caused by grounds to the sheath.

3.14.3 Number of defective type II pairs.- The number of defective type II pairs caused by shorts, opens or crosses shall not be more than permitted as specified in Table VII. Each type II pair which is found defective shall be marked with linen tags, colored metal beads, or hard rubber sleeves at each end of the length of cable with the nature of the defect indicated.

TABLE VII

DEFECTIVE TYPE II PAIRS

Number of Type II Pairs in Cable	Allowable number of Defective Type II Pairs
1-8 pairs	0
1-125 pairs	1
1-430 pairs	2

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3.15 Gas filled cable.- All cables on reels shall have nitrogen gas within the lead sheath. The cable ends shall be sealed with a solder having a tin content of not less than 40 percent. A standard pneumatic tire valve shall be inserted in the center of each cable end. The solder shall make a gastight bond with the valve under the pressure condition as specified in 4.9. A lead cap, painted yellow, shall be placed over the valve so as to form a watertight joint and prevent the valve from becoming loose in shipment. Both ends of every length of cable, containing defective pairs, shall be painted red for a minimum of 4 inches. The pressure and temperature after the tests, and the date of the test, shall be recorded on a durable tag secured to the outer end of the cable.

3.16 Workmanship.- All cable shall be manufactured and finished in a workmanlike manner in accordance with QQ-W-343.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. (See 6.2.d)

4.1.1 Classification of tests.- The inspection and testing of the cable shall be as follows.

4.1.2 Test conditions.- Unless otherwise specified, all tests required by this specification shall be made at prevailing ambient temperature and humidity conditions.

4.1.3 Inspection lot.- A lot shall consist of all products of one type, produced under substantially the same conditions on a continuous basis, and offered for inspection at one time. All constructions of Class 1 cables shall be considered one type and all constructions of Class 2 cables shall be considered another type.

4.1.4 Lot size.- The lot size is the number of units of product in an inspection lot. The minimum and maximum lot sizes for inspection purposes shall be in accordance with MIL-STD-105 and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

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4.1.5 MIL-STD-105.- When MIL-STD-105 specifies an action by the Government, it shall at the option of the Government, be performed by the Government or by the contractor under the supervision of the Government Inspector.

4.2 Acceptance inspection.- Acceptance inspection shall consist of group A, group B, and group C inspection as specified in 4.2.1 through 4.2.3.2. Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.2.1 Group A inspection.- Group A inspection including sampling shall be in accordance with Table VIII and MIL-STD-105.

TABLE VIII

GROUP A INSPECTION

Inspection	Required Paragraph	Test Paragraph	Acceptable Quality Level (AQL)
Visual and mechanical (major)	--	4.3	1 percent
Visual and mechanical (minor)	--	4.3	4 percent
Electrical:			
Dielectric strength			
Type I pairs	3.13.1.6	4.7.1.5	1 percent
Type II pairs	3.13.2.6	4.7.2.5	1 percent
Insulation resistance			
Type I pairs	3.13.1.3	4.7.1.8	1 percent
Type II pairs	3.13.2.3	4.7.2.8	1 percent
Continuity of conductors			
Type I pairs	3.13.1.1	4.7.1.2	1 percent
Type II pairs	3.13.2.1	4.7.2.2	1 percent
Conductor resistance			
Type I pairs	3.13.1.2	4.7.1.1	1 percent
Type II pairs	3.13.2.2	4.7.2.1	1 percent
Capacitance			
Type I pairs	3.13.1.4	4.7.1.3	1 percent
Type II pairs	3.13.2.4	4.7.2.3	1 percent
Capacitance unbalance			
Type I pairs	3.13.1.5	4.7.1.4	1 percent
Type II pairs	3.13.2.5	4.7.2.4	1 percent
Echo			
Type I pairs	3.13.1.7	4.7.1.6	1 percent
Type II pairs	3.13.2.7	4.7.2.6	1 percent

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TABLE VIII

GROUP A INSPECTION

Inspection	Required Paragraph	Test Paragraph	Acceptable Quality Level (AQL)
Electrical:			
Impedance			
Type I pairs	3.13.1.8	4.7.1.7	1 percent
Type II pairs	3.13.2.8	4.7.2.7	1 percent
Defective pairs	3.14	--	1 percent
Gas test	3.15	4.9	1 percent

NOTE: The total number of defective pairs in the lot, regardless of the defect, shall be used to determine acceptance or rejection of the lot.

4.2.1.1 Unmarked defective pairs.- Unmarked defective pairs found as a result of inspection shall be a basis of rejection of the lot if the number of defective pairs found exceeds the acceptance number specified in MIL-STD-105, even though the resulting total number of defective pairs in any one cable is within the tolerances permitted in Table VII.

4.2.2 Group B inspection.- Group B inspection, including sampling, shall conform to Table IX and MIL-STD-105. The acceptable quality level shall be 6.5 percent and the inspection level shall be S-4 for normal inspection and S-3 for reduced inspection. Unless otherwise specified herein, normal inspection shall be used at the start of the contract.

TABLE IX

GROUP B INSPECTION

Inspection	Requirement Paragraph	Test Paragraph
Insulation of conductors	3.3	4.5
Insulating paper	3.3.2.1	4.6
Conductor size		
Type I conductor	3.2.1	4.8
Type II conductor	3.2.2	4.8
Core belting paper	3.3.4	
Heat damage	3.4	4.8
Color coding		
Type I pairs	3.5.1.3	4.8
Type II pairs	3.5.2	4.8
Cabling	3.6	4.8
Thickness of sheath	3.7.1	4.8
O.D. of cable	3.11	4.8

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4.2.2.1 Procedure in case of failure.- When a lot is rejected, acceptance of all items in the lot shall be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.2.3 Group C inspection.- Group C inspection shall be as specified in Table X.

TABLE X

GROUP C INSPECTION

Inspection	Requirement Paragraph	Test Paragraph
Joints	3.2.3	4.4
Softening point	3.8.4.1	4.10
Ductility	3.8.4.2	4.11

4.2.3.1 Samples

4.2.3.1.1 Joints.- Each person fabricating joints in the cable conductors, shall prepare one sample joint per month. These joints shall be equivalent to those in the finished cable using the same equipment and techniques. The joints shall be made on a length of wire similar to that used in the finished cable.

4.2.3.1.2 Softening point and ductility.- Samples of the flooding compound shall be taken from each saturating tank at least once for each lot of cable.

4.2.3.2 Noncompliance.- If a sample joint fails group C inspection, the contractor shall immediately investigate the cause of failure and shall take corrective action. If the cause of failure was poor workmanship on the part of the operator, the operator in question shall not be permitted to make production joints until he has made six additional joints, using the proper procedures, which pass the group C inspection. If a sample fails the group C softening point or ductility test, the contractor shall immediately investigate the cause of failure and shall take corrective action. The sampling shall be increased on the failing test until such time that the cause of the failure has been eliminated.

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4.3 Visual and mechanical inspection.- The cable, pneumatic valves, and cable reels shall be examined for workmanship, mechanical fit, and miscellaneous defects. Labels and markings shall be inspected for conformance with approved facsimiles when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified as major or minor in accordance with the definitions of MIL-STD-105.

4.4 Conductor joint test

4.4.1 Tensile strength.- The tensile strength test shall be made in accordance with method 3021 of J-C-98. The following additional tensile tests shall also be made:

a. Measure tensile strength of cable sheath using samples of 0.04 to 0.08 percent of copper bearing lead.

b. Make the same test for the same size sample of 1 percent antimony lead.

c. Record readings and compute the tensile strength of each sample in pounds per square inch. A minimum of three samples of each should be tested and recorded, then the average tensile strength of each two lead composites computed.

d. The tensile strength and maximum allowable pulling tension of the two complete cable samples should be determined. Tests should be made with the cable samples equipped with pulling eyes at each end. The elongation and maximum tension before rupture should be recorded for each test. Test should be repeated using commercial cable grips.

4.4.2 DC resistance.- The DC resistance shall be made in accordance with method 6021 of J-C-98 except that the accuracy of the bridge shall be within 0.3 percent.

4.5 Insulation.- The insulation on the conductors shall be given a visual inspection for compliance with 3.3.

4.6 Insulating paper and belting paper.- The contractor shall submit certified, recently performed data obtained by tests on insulating paper of the type used to manufacture the cable on order, to determine compliance with the requirements of 3.3.2.1 and 3.3.4.

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4.7 Electrical tests

4.7.1 Type I pairs

4.7.1.1 Conductor resistance.- The conductors of a length of finished cable shall be tested for resistance to meet the requirements specified in 3.13.1.2. The test shall be performed in accordance with method 6021 of J-C-98.

4.7.1.2 Conductor continuity.- The continuity of the conductors in a length of cable shall be tested with a low DC voltage and a continuity indicator. (See 3.13.1.1)

4.7.1.3 Capacitance.- The mutual capacitance of a pair shall be measured at a frequency of either 900 or 1000 CPS between two wires of a pair, the shields of the type I pairs connected to the sheath and grounded. The capacitance shall meet the requirements specified in 3.13.1.4 for completed cable less than 300 feet in length.

4.7.1.4 Capacitance unbalance.- The capacitance unbalance shall be measured at a frequency of either 900 or 1000 CPS. Capacitance unbalance shall be the difference in the direct capacitance of one conductor of a pair to shield and the other conductor of a pair to shield as measured in the finished cable and shall meet the requirements specified in 3.13.1.5.

4.7.1.5 Dielectric strength.- A 60 cycle AC potential whose maximum instantaneous value is not less than that specified shall be applied for two seconds. The insulation between the conductors, between conductor and shields and between shield and sheath, shall withstand the potentials specified in 3.13.1.6.

4.7.1.6 Internal echo.- The magnitude of the internal echo of each end of each pair in a length of cable 300 feet and longer shall meet the requirements specified in 3.13.1.7. The magnitude of an internal echo shall be considered as the voltage ratio in DB between the peak voltage of the incident wave and the peak voltage of the positive or negative reflection at the point of reflection. Allowance shall be made for the attenuation of the pulse to the point of reflection. The incident wave shall be a sharp pulse of one megacycle (MC) frequency, which rises sharply from zero and returns sharply to zero with a width of 1/4 microseconds (USEC) midway between the base and the peak. The over-all reflection coefficient of the equipment shall be not less than 0.003.

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4.7.1.7 Terminal impedance.- The magnitude of the terminal impedance of each pair in a length of cable 300 feet and longer, shall meet the requirements specified in 3.13.1.8. The terminal impedance is considered as 1 MC impedance of a network having approximately the same impedance frequency characteristics as the type I pairs such that when connected to the type I pair on the test, it produces a junction echo at least 43 DB down, with the transmission of a sharp pulse.

4.7.1.8 Insulation resistance.- The insulation resistance of the conductors shall be determined by measuring the insulation resistance of each individual conductor against all other conductors grounded to the sheath. The insulation resistance shall meet the requirements specified in 3.13.1.3.

4.7.1.8.1 Group testing.- Group testing for insulation resistance may be employed. A small number of similarly colored conductors (not more than ten conductors) may be connected together and the combined insulation resistance of the group measured against all other conductors grounded to the sheath. As an alternative to connecting together a small number of similarly colored conductors and measuring to all other conductors, the conductors may be so grouped that the insulation resistance is measured between each conductor and its contiguous conductors and to the sheath.

4.7.1.8.2 Retesting.- If the cable, when measured at a temperature higher than 15°C (59°F), fails to meet the requirements of 3.13.1.3, no temperature correction factor shall be applied, but in such cases the cable may be retested at a temperature not lower than 15°C (59°F).

4.7.2 Type II pairs

4.7.2.1 Conductor resistance.- Select for inspection 22 conductors or 10 percent of the conductors, whichever is greater. The conductors of a length of cable shall be selected throughout the cross section of the cable and shall be tested for resistance of conductors as specified in 3.13.2.2.

4.7.2.2 Conductor continuity.- The continuity of the conductors in a length of cable shall be tested with a low DC voltage and a continuity indicator. (See 3.13.2.1)

4.7.2.3 Capacitance.- The mutual capacitance of a pair shall be measured between the two wires of the pair, the remainder of the conductors being connected to the sheath which shall be grounded. The capacitance shall meet the requirements specified in 3.13.2.4.

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4.7.2.3.1 Retesting.- If the cable, when measured at a temperature of higher than 15°C (59°F), fails to meet the requirements of 3.13.2.4, no temperature correction factor shall be applied, but in such cases the cable may be retested at a temperature not lower than 15°C (59°F).

4.7.2.4 Capacitance unbalance.- The capacitance unbalance shall be measured at a frequency of either 900 or 1000 CPS. The capacitance unbalance shall be the difference in the direct capacitance of one conductor of a pair to shield and the other conductor of a pair to shield as measured in the finished cable and shall meet the requirements specified in 3.13.2.5.

4.7.2.5 Dielectric strength.- The conductors in a length of cable shall be subjected to a dielectric strength test both between the conductor, and the remaining conductors connected together and between the conductor, sheath and shields of the type II pairs using the potentials specified in 3.13.2.6 and 3.13.2.6.1.

4.7.2.5.1 Group testing.- Group testing for dielectric strength may be employed. The conductors may be grouped so that the voltage applied shall be between each conductor and its contiguous conductor and to the sheath.

4.7.2.5.2 Alternate test procedure.- The alternate test procedure as specified in table XI may be used in place of the test procedure specified in 4.7.2.5.

a. Arrange the conductors in a rectangle containing rows and columns as specified in table XI. The last column will contain unused spaces as specified.

TABLE XI

ALTERNATE DIELECTRIC STRENGTH TESTS

Number of Conductors	Number Rows	Number Columns	Number Unused Spaces
22	5	5	3
32	6	6	4
52	7	8	4
102	10	11	8
152	12	13	4
202	14	15	8
304	17	18	2
404	20	21	16
506	22	23	0
606	25	25	19
808	28	29	4
1212	35	35	13
1818	43	43	31

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b. Connect all the conductors in a column together. Do this for each column.

c. Apply the specified test voltage, for the specified time, between each column of the rectangle in turn, and the remaining columns connected together.

d. Connect all the conductors in a row together. Do this for each row.

e. Apply the specified test voltage, for the specified time, between each row of the rectangle in turn, and the remaining rows connected together.

f. Connect all the conductors together.

g. Apply the specified voltage, for the specified time, between all the conductors connected together and the sheath.

4.7.2.6 Internal echo.- The magnitude of the internal echo of each end of each pair in a length of cable 300 feet and longer shall meet the requirements specified in 3.13.2.7. The magnitude of an internal echo shall be considered as the voltage ratio in DB between the peak voltage of the incident wave and the peak voltage of the positive or negative reflection at the point of reflection. Allowance shall be made for the attenuation of the pulse to the point of reflection. The incident wave shall be a sharp pulse of one MC frequency, which rises sharply from zero and returns sharply to zero with a width of 1/4 USEC midway between the base and the peak.

4.7.2.7 Terminal impedance.- The magnitude of the terminal impedance of each pair in a length of cable 300 feet and longer shall meet the requirements specified in 3.13.2.8. The terminal impedance is considered as 1 MC impedance of a network having approximately the same impedance frequency characteristics as the type I pairs such that, when connected to the type I pair on the test, it produces a junction echo at least 43 DB down with the transmission of a sharp pulse.

4.7.2.8 Insulation resistance.- The insulation resistance of the conductors may be determined by measuring the insulation resistance of each individual conductor against all other conductors grounded to the sheath. The insulation resistance shall meet the requirements specified in 3.13.2.3.

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4.7.2.8.1 Group testing.- Group testing for insulation resistance may be employed. A small number of similarly colored conductors (not more than ten conductors) may be connected together and the combined insulation resistance of the group measured against all other conductors grounded to the sheath. As an alternative to connecting together a small number of similarly colored conductors and measuring to all other conductors, the conductors may be so grouped that the insulation resistance is measured between each conductor and its contiguous conductors and to the sheath.

4.7.2.8.2 Retesting.- If the cable, when measured at a temperature higher than 15°C (59°F), fails to meet the requirements of 3.13.2.3, no temperature correction factor shall be applied, but in such cases the cable may be retested at a temperature not lower than 15°C (59°F).

4.8 Cable make-up.- Specimens of cable from each sample unit shall be inspected for dimensions, assembly order and color coding as required in 3.2, 3.4, 3.5, 3.6, 3.7.1 and 3.11.

4.9 Gas test.- The lead sheath and seals of all cables shall be tested for possible defects by introducing dry nitrogen or dry air. The gauge pressure at both ends of the cable, after the gas has been stablized, shall be between 11 and 13 PSI at a temperature of 15°C (59°F). At the end of 12 hours, the pressure, after the correction for any temperature change is applied, shall not be less than the stabilized value of between 11 and 13 PSI. The cable shall remain under pressure for delivery.

4.10 Softening point test.- The method of the softening point test shall be in accordance with Standard, American Society For Testing Materials, (ASTM), Methods of Test For Softening Point of Bituminous Material (Ring and Ball Method), Designation D-36, latest issue.

4.11 Ductility test.- The method of test shall be in accordance with Standard (ASTM), Methods of Test For Ductility of Bituminous Materials. Designation D-113, latest issue.

5. PREPARATION FOR DELIVERY

5.1 The cable specified herein shall be prepared and marked for shipment in accordance with MIL-C-12000. (See 6.2.c)

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6. NOTES

6.1 Intended use.- Combination telephone and video pair cable having a plain lead covered sheath or buried tape armored protection over the lead sheath, as covered by this specification, is intended for transmitting electrical impulses. The plain lead covered cables are suitable for aerial or duct installations, and the tape armored cables are suitable for direct earth burial. As the construction of the video pair consists of a low melting insulation, precautions should be taken to install the cables at locations where the temperatures do not exceed 79.5°C (175°F) for an extended length of time.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Class and designation of cable to be furnished. (See 1.2)

c. Applicable level of preservation, packaging and packing required. (See 5.1)

d. Whether the testing will be conducted by the procuring activity or by the contractor. (See 4.1)

6.3 Definitions.- For the purpose of this specification definitions given below shall apply.

6.3.1 Unit of product.- The unit of product is a continuous length of finished cable contained on a reel.

6.3.2 Sample.- A sample is one or more sample units selected from an inspection lot to represent that inspection lot for inspection.

6.3.3 Sample unit.- A sample unit is a unit of product, selected in an unbiased manner to be part of a sample, from which specimens are taken for inspection.

6.3.4 Specimen.- A specimen is a single piece of finished cable which is taken from a sample unit and subjected to inspection.

6.3.5 Finished cable.- Finished cable is cable on which all manufacturing operation have been completed and which is ready to be submitted for acceptance.

Custodian:
Air Force - 17

Preparing activity:
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